

AWS Lambda deployment

IAM

>

Roles

>

Create role

Step 1

●

Select trusted entity

Step 2

●

Add permissions

Step 3

●

Name, review, and create

Select trusted entity

Info

Trusted entity type

☒ AWS service

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ AWS account

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ Web identity

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ SAML 2.0 federation

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Lambda

Choose a use case for the specified service.

Use case

☒ Lambda

Allows Lambda functions to call AWS services on your behalf.

☐ AWSServiceRoleForLambda

Allows Lambda to manage AWS resources on your behalf.

AWSLambdaBasicExecutionRole

Add permissions

Info

Permissions policies (1/1140)

Info

Choose one or more policies to attach to your new role.

Filter by Type

Search AWSLambdaBasicExecutionRole

All types

Policy name

Type

☒

[AWSLambdaBasicExecutionRole](#)

AWS managed

☐

[AWSLambdaBasicExecutionRole-8e55c59a-9c24-402f-be18-2d2b38671518](#)

Customer managed

☐

[AWSLambdaBasicExecutionRole-aa009388-c26a-4337-a34d-3e46bf7c22fc](#)

Customer managed

► Set permissions boundary - optional

Role details

Role name

Enter a meaningful name to identify this role.

lambda-execution-role

Maximum 64 characters. Use alphanumeric and '+,=,@-_' characters.

Description

Add a short explanation for this role.

Allows Lambda functions to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=, @-/\[\]!#

Step 1: Select trusted entities

Trust policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "sts:AssumeRole"
8       ],
9       "Principal": {
10        "Service": [
11          "lambda.amazonaws.com"
12        ]
13      }
14    ]
15  }
16 }
```

Step 2: Add permissions

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "sts:AssumeRole"
      ],
      "Principal": {
        "Service": [
          "lambda.amazonaws.com"
        ]
      }
    }
  ]
}
```

Copy role ARN and save for auto-deployment

Create Lambda

Basic information

Function name
Enter a name that describes the purpose of your function.
GraphQL-FastAPI
Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.
Python 3.10

Durable execution - new [Info](#)
Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#).
☐ Enable

Architecture [Info](#)
Choose the instruction set architecture you want for your function code.
☐ arm64
☒ x86_64

Permissions [Info](#)
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ **Change default execution role**

Execution role
To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

☐ Create default role ☒ Use another role

lambda-execution-role [View role details in IAM](#)

[Create new role](#)

Assign IAM role we just created and Create Function
Upload file

Upload a .zip file

When you upload a new .zip file package, it overwrites the existing code.

Upload

deployment.zip

10.37 MB

For files larger than 10 MB, consider uploading using Amazon S3.

Encryption with an AWS KMS customer managed key

☐

By default, Lambda encrypts the .zip file archive using an AWS owned key.

Cancel

Save

Runtime settings

Runtime settings

Runtime

Python 3.10

Handler

lambda_function.lambda_handler

Architecture

x86_64

▼ Runtime management configuration

Runtime version ARN

arn:aws:lambda:us-east-1::runtime:3c9d8d4e1f6ed9432c45c88367aa212b84d8809f294df39c002d4f3599c8afb

Update runtime version

Auto

Edit

Edit runtime management configuration

Edit => main.handler

Edit runtime settings

Runtime settings

Runtime

Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.10

A new runtime is available for your function's language: Python 3.14

Handler

main.handler

You can change either the function's runtime or the instruction set architecture in one update. To update both, you must repeat the update process.

Architecture

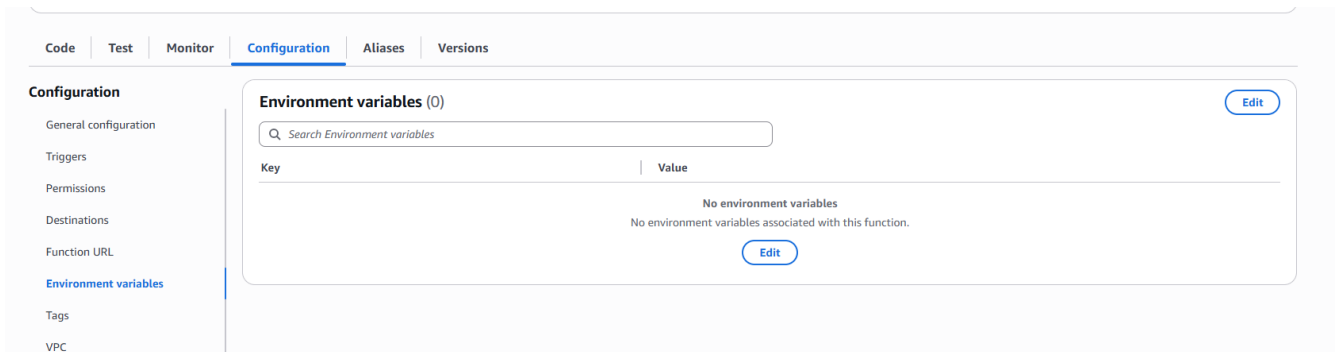
Choose the instruction set architecture you want for your function code.

arm64

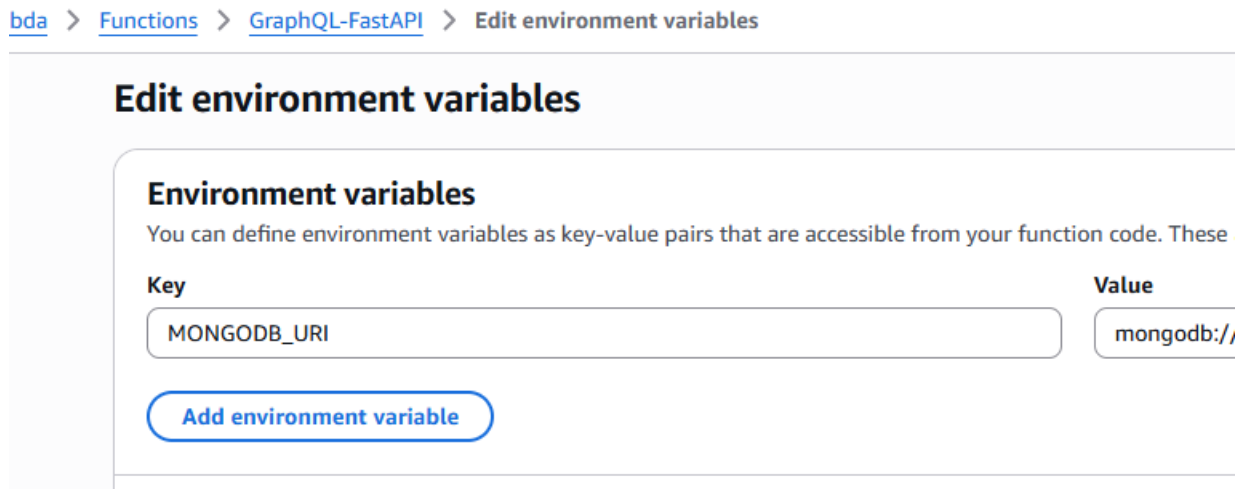
x86_64

Cancel

Save

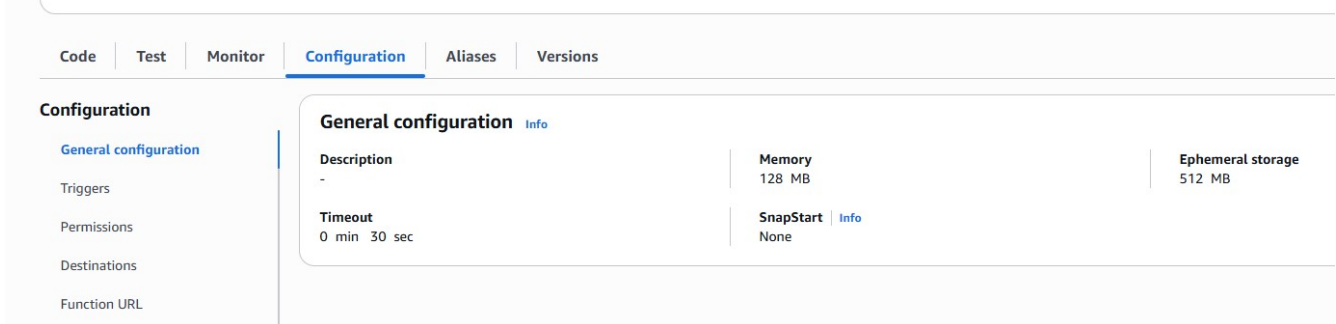


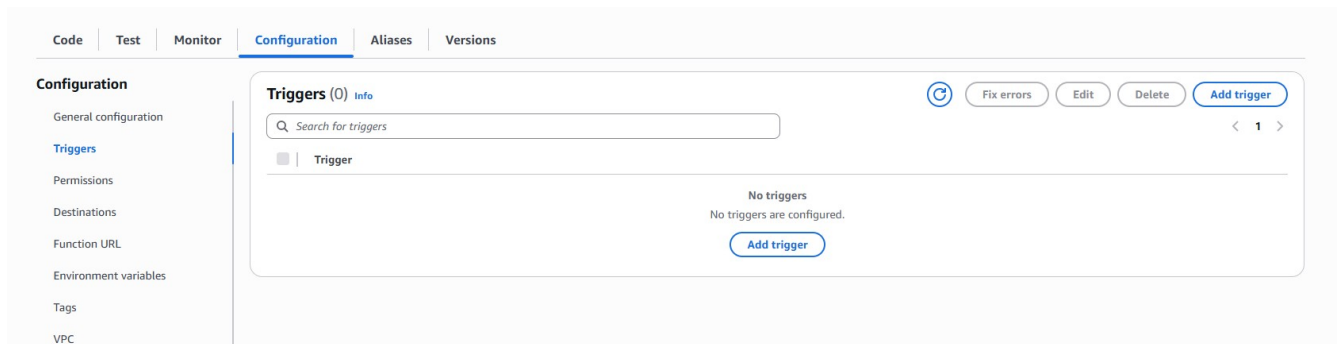
Add Environment variables: MONGODB_URI:



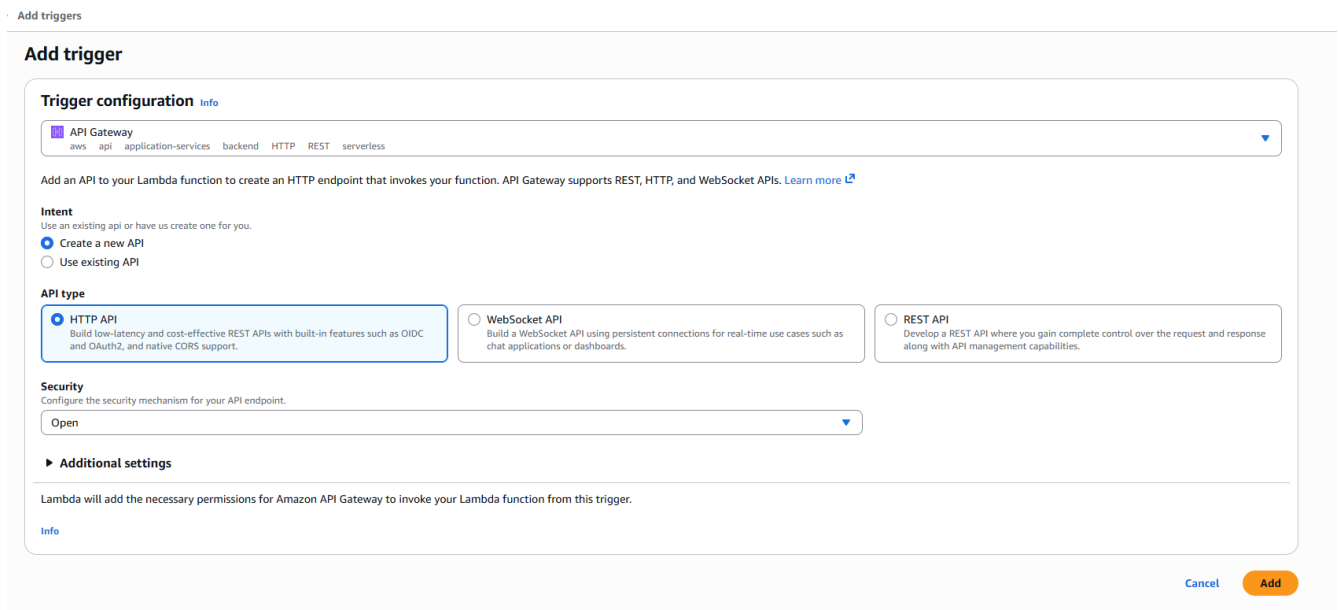
Save

Configuration => General configuration => Timeout 30 sec





Add trigger



This is the GraphQL URL

<https://m6nolf37md.execute-api.us-east-1.amazonaws.com/default/GraphQL-FastAPI>



Go to Test page

```
{
  "version": "1.0",
  "httpMethod": "GET",
  "path": "/GraphQL-FastAPI/graphql",
  "headers": {},
  "queryStringParameters": null,
  "body": null,
  "isBase64Encoded": false,
  "requestContext": { "stage": "default" }
}
```

CodeTestMonitorConfigurationAliasesVersions

✔ Executing function: succeeded (logs 12)

▼ Details

```
{
  "statusCode": 400,
  "headers": {
    "content-length": "37",
    "content-type": "text/plain; charset=utf-8"
  },
  "multiValueHeaders": {},
  "body": "No GraphQL query found in the request",
  "isBase64Encoded": false
}
```

Summary

Code SHA-256 rWAgyiux/uF3OA0C5/fvdfzNxUHgg397hyaxuTvIGoY=	Execution time 3 seconds ago
Function version \$LATEST	Request ID ab223808-4169-4d59-8588-4d5e7b603acb
Duration 10.47 ms	Billed duration 2344 ms
Resources configured	Max memory used

0
0
3
0
0

Stage: -
Deploy

Route details

Delete
Edit

ANY /GraphQL-FastAPI (ID: 1tkgiwe)

ARN
arn:aws:apigateway:us-east-1::apis/m6nolf37md/routes/1tkgiwe

Authorization
Authorizers protect your API against unauthorized requests. Routes with no authorization attached are open.

No authorizer attached to this route. [Attach authorization](#)

Integration
The integration is the backend resource that this route calls when it receives a request.

h31amdu [Configure](#)

Click Deploy

Make sure we include 0.0.0.0/0 in MongoDB Network access

Successfully updated the function GraphQL-FastAPI

Code
Test
Monitor
Configuration
Aliases
Versions

Executing function: succeeded (logs)

Details

```

{
  "statusCode": 200,
  "headers": {
    "content-length": "192",
    "content-type": "application/json"
  },
  "multiValueHeaders": {},
  "body": "{\"query\": \"{ users { id name age phone } }\"\", \"isBase64Encoded\": false"
}

```

Summary

```

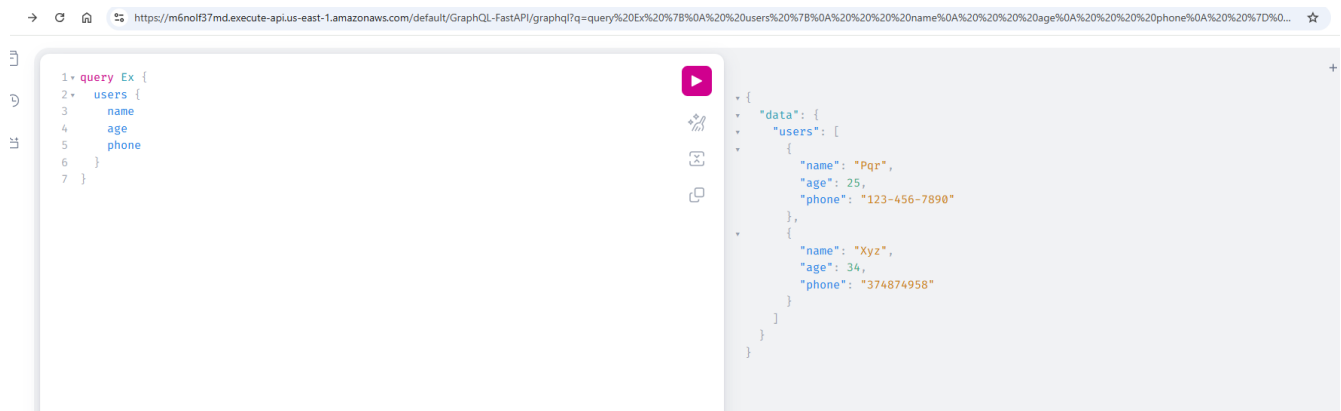
{
  "version": "1.0",
  "httpMethod": "POST",
  "path": "/GraphQL-FastAPI/graphql",
  "headers": {
    "content-type": "application/json"
  },
  "queryStringParameters": null,
  "body": "{\"query\": \"{ users { id name age phone } }\"\", \"isBase64Encoded\": false",
  "requestContext": {
    "stage": "default",
    "identity": { "sourceIp": "127.0.0.1" }
  }
}

```


Deployed on Lambda

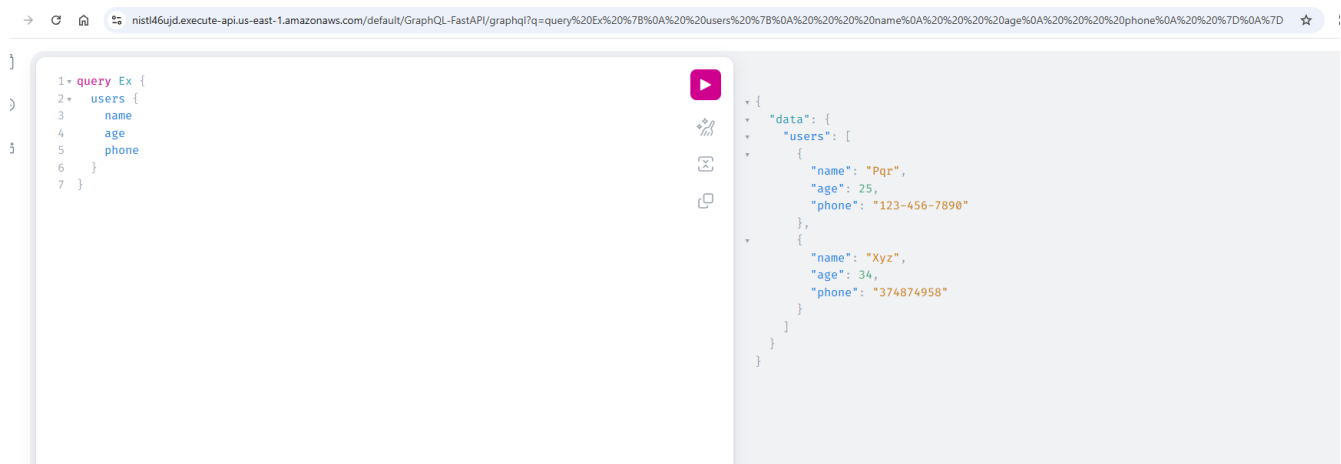
<https://m6nolf37md.execute-api.us-east-1.amazonaws.com/default/GraphQL-FastAPI/graphql?q=query%20Ex%20%7B%0A%20%20users%20%20%7B%0A%20%20%20%20name%0A%20%20%20%20age%0A%20%20%20%20phone%0A%20%20%7D%0A%7D>

<https://m6nolf37md.execute-api.us-east-1.amazonaws.com/default/GraphQL-FastAPI/graphql>



Tested auto-deployment: <https://nistl46ujd.execute-api.us-east-1.amazonaws.com/default/GraphQL-FastAPI/graphql>

<https://nistl46ujd.execute-api.us-east-1.amazonaws.com/default/GraphQL-FastAPI/graphql?q=query%20Ex%20%7B%0A%20%20users%20%20%7B%0A%20%20%20%20name%0A%20%20%20%20age%0A%20%20%20%20phone%0A%20%20%7D%0A%7D>



```
query Ex {  
  users {  
    name  
    age  
    phone  
  }  
}
```